SAHAYAK - PRESENTATION GUIDE

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SYSTEM DESIGN, TECHNOLOGY AND WORKFLOW

Suggested speaking time: 4-5 minutes

Speaking script, slide cues, handover, troubleshooting and judge Q&A

YOUR ROLE

You are the technical speaker. Explain the data path step by step: camera, MediaPipe hand landmarks, feature vector, backend model, predicted label, website display. The judges do not need every line of code; they need to understand what each component contributes.

SLIDE 6 - THE COMPLETE WORKFLOW

Say: Thank you. I will explain what happens behind the screen. First, the browser requests access to the camera. MediaPipe Hands detects hand landmarks. The application converts the detected points into numerical features and sends them to the Python Flask backend. A trained model predicts a label, and the website displays the result.

The browser handles the camera preview and landmark overlay; the backend handles model inference. This separation keeps the visual interface and model processing independent.

SLIDE 7 - WHAT ARE HAND LANDMARKS?

Say: A landmark is a reference point on the hand, such as a fingertip or joint. MediaPipe detects 21 landmarks for each hand. Each landmark is represented by x, y, and z coordinates. The numbers describe hand shape and position in a form that a machine-learning model can process.

Simple example: A person may look at a hand and notice a bent finger. The feature vector represents landmark positions numerically. The model learns statistical patterns from labelled examples; it does not understand signs like a human signer.

SLIDE 8 - STATIC FEATURES: WHY 126 VALUES?

Say: Each hand has 21 landmarks, and each landmark has three coordinates. So one hand gives 21 times 3, which equals 63 values. For two hands, 63 times 2 equals 126 values. That is the static feature vector used by the two-hand prediction pathway.

Write: 2 hands x 21 landmarks x 3 coordinates = 126 values.

The current intended label set includes HELLO, YES, NO, THANK YOU, and I LOVE YOU. Actual recognition is limited by what the deployed model was trained to recognise and how it performs. A label in a file alone does not prove reliable recognition.

SLIDE 9 - WHY SEQUENCES MATTER

Say: Some signs involve movement, so a single frame may not capture the full pattern. SAHAYAK also has a sequence pathway. Each frame uses 132 values, including 126 hand-landmark coordinates plus six wrist-position values. A sequence is resampled into 10 time bins, giving 132 times 10, or 1,320 values for the sequence model.

This is the designed input format. It does not prove every moving sign works. We need labelled sequence samples and fair evaluation to show which continuous signs the deployed

model recognises correctly.

SLIDE 10 - FRONTEND, BACKEND AND MODEL

Say: The frontend website is hosted on Vercel, and the Python Flask backend is hosted on Render. The frontend sends prediction requests to backend API routes. The backend loads model files and returns prediction results. The browser draws the camera view and hand landmarks.

The current backend uses scikit-learn/joblib model artifacts. Do not describe it as a TensorFlow or convolutional-neural-network system unless the implementation changes and is verified. Some online translation or speech functions need internet access, so do not claim the entire system is fully offline.

SLIDE 11 - DATA, TRAINING AND FAIR EVALUATION

Say: To teach a model a new sign, we need correctly labelled examples. Then we check the samples, train a model with a compatible feature format, and test it on examples not used for training. Ideally, test data should include different people or recording sessions to reduce the risk of memorising familiar examples.

Accuracy is not the same as response speed, and feature count is not dataset size. We should not quote an accuracy percentage until we have documented testing results. Useful evidence includes results per sign, confusion between labels, and trials across lighting and camera conditions.

HANDOVER TO HIMANXHU V GHIMIRE

Say: That is the technical workflow from camera input to prediction. My teammate Himanxhu V Ghimire will now demonstrate the website, explain how to test it fairly, and discuss limitations and future improvements.

JUDGE QUESTIONS AND ANSWERS

Q: Why use landmarks instead of the whole image? A: Landmarks give a compact numerical description of hand shape and position.

Q: What does 126 mean? A: Two hands times 21 landmarks times three coordinates per landmark.

Q: Static versus sequence recognition? A: Static uses one frame’s features; sequence recognition uses information across time.

Q: Does adding a label automatically add a sign? A: No. We need labelled data, model training, evaluation, and deployment.

Q: Where is recognition calculated? A: The browser detects and annotates landmarks; the trained classifier is loaded by the backend for prediction.

TECHNICAL CHECKLIST

Remember: 126 static values; 132 values per sequence frame; 10 time bins; 1,320 sequence input values. Distinguish architecture from measured performance. If the live request fails, explain the workflow using the diagram instead of pretending a prediction succeeded.